

Diagonalizability & the minimal polynomial

Proposition : $A \in M_{n \times n}(\mathbb{C})$

TFAE: (1) A is diagonalizable

(2) $\mathbb{C}^n$ admits an eigenbasis for A

(3) $\min_A(x)$ has no repeated linear factors

i.e. $\min_A(x) = \prod_{i=1}^r (x - \lambda_i)$
for $\lambda_1, \dots, \lambda_r \in \mathbb{C}$ distinct

Need to know

(1) $\iff$ (3)

Observation ① $A \in M_{n \times n}(F)$

$$f(x) \in \text{Poly}(F) \text{ s.t. } f(A) = 0$$

If $C \in M_{n \times n}(F)$ invertible

$$\text{then } f(\underbrace{CAC^{-1}})$$

Conjugate of A by C

Pf. $f(x) = a_m x^m + \dots + a_1 x + a_0$

$\downarrow$ given
 $0 = f(A) = a_m A^m + \dots + a_1 A + a_0 \cdot I_n$

$$f(CAC^{-1}) = a_m (CAC^{-1})^m + \dots + a_1 (CAC^{-1}) + a_0 \cdot I_n$$

$$= C(a_m A^m)C^{-1} + \dots + C(a_1 A)C^{-1} + C(a_0 I_n)C^{-1}$$

$$= C[a_m A^m + \dots + a_1 A + a_0]C^{-1}$$

$$= 0.$$

Observation (2)

Notation as above

$$\left\{ \begin{array}{l} f(x) \in \text{Poly}(F) : \\ f(A) = 0 \end{array} \right\} = \left\{ \begin{array}{l} f(x) \in \text{Poly}(F) : \\ f(CAC^{-1}) = 0 \end{array} \right\}$$

pf: Obs. (1) $\Rightarrow$ LHS $\subseteq$ RHS

Since $A = C^{-1}(CAC^{-1})C$

the same observation tells me
RHS $\subseteq$ LHS.

Observation (3)

$$\min_A(x) = \min_{CAC^{-1}}(x)$$

That is, the minimal polynomial

is basis independent

Pf of Proposition

(1)

||

A is diagonalizable



(3)

||

$\min_A(x)$ has
no repeated factors

By Obs. (3), we can assume

$$\text{that } A = \begin{bmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \lambda_n \end{bmatrix}$$

is diagonal

Observation (4) : $(A - \lambda_i I_n) e_i$

$$= A e_i - \lambda_i e_i = \lambda_i e_i - \lambda_i e_i = 0.$$

$$\lambda_i \neq \lambda_j$$

$$(A - \lambda_j I_n)(A - \lambda_i I_n)e_i \\ = 0$$

$$(A - \lambda_j I_n)(A - \lambda_i I_n)e_i \\ = (A - \lambda_j I_n)(\lambda_i - \lambda_i)e_i \\ = (\lambda_i - \lambda_i)(A - \lambda_j I_n)e_i \\ = 0.$$

This argument tells us that if we take

$$f(x) = \prod_{\lambda_i \text{ distinct}} (x - \lambda_i)$$

then

$$f(A) \cdot e_k = 0, \quad \forall k$$

$$\Rightarrow f(A) = 0$$

$$\Rightarrow f(x) = \min_A(x) h(x)$$

$\Rightarrow f(x)$ has no repeated factors
and hence $\min_A(x)$ also has
no repeated factors

Rank:

$$B = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix} ; \quad \text{ch}_B(x) = (x-2)^2$$

$$\min_B(x) = (x-2)^2$$

$\Rightarrow B$ is non-diagonalizable

(3)

$\Rightarrow$

(1)

$\uparrow$
 $\min_A(x)$

has no
repeated
factors

$\uparrow$

A is diagonalizable

Defⁿ (1) $A \in M_{n \times n}(\mathbb{C})$

$$\text{Ch}_A(x) = (-1)^n \prod_{\lambda \text{ distinct}} (x - \lambda)^{m(\lambda)}$$

$m(\lambda)$: algebraic multiplicity
of λ -value λ .

$$\begin{aligned} (2) \quad E_\lambda &= \ker(A - \lambda I_n) \\ &= \{v \in \mathbb{C}^n : Av = \lambda v\} \\ &\uparrow \\ &\lambda\text{-eigenspace for } A \end{aligned}$$

$$\begin{aligned} (3) \quad &\text{geometric multiplicity of } \lambda \\ &= \dim E_\lambda \end{aligned}$$

Rmk: TFAE:

- (1) λ is an eigenvalue for A
- (2) $m(\lambda) \neq 0$
- (3) $\dim E_\lambda \neq 0$

Fact: $A \in M_{n \times n}(\mathbb{C})$

A is diagonalizable

$$(\Leftarrow) \quad \dim E_2 = m(2)$$

$$\forall \lambda \in \mathbb{C}.$$

Pf.: A is diagonal

A handwritten diagram on lined paper showing a large square matrix structure enclosed in brackets. The matrix is block diagonal, consisting of several square blocks along the main diagonal and zeros elsewhere. The blocks are labeled as follows:

- The top-left block is labeled $m(h_1)$ and has a vertical line above it with the number 11.
- The second block is labeled $m(h_2)$.
- The third block is labeled $m(h_r)$.
- The bottom-right block is labeled $m(h_r)$.

 Between the blocks, there are labels h_1, h_2, h_r and ellipses indicating a sequence of blocks. There are also two circles, one in the top right and one in the bottom left of the matrix structure, possibly representing zero blocks or specific elements.

$$E_{\lambda_1} = A - \lambda_1 I_n = \begin{bmatrix} 0 & & & \\ & \ddots & & \\ & & \lambda_2 - \lambda_1 & \\ & & \ddots & \ddots \\ 0 & & & \lambda_r - \lambda_1 \end{bmatrix}$$

$$\Rightarrow \dim E_{\lambda_i} = m(\lambda_i)$$

Conversely, if $\dim E_{\lambda_i} = m(\lambda_i), \forall i$

then the claim is we have

an eigenbasis.

E_{λ_1}	E_{λ_1}	$\dots$	E_{λ_r}
$\uparrow$ basis	$\uparrow$		
$(v_{1,1}, \dots, v_{1,m(\lambda_1)})$	$(v_{2,1}, \dots, v_{2,m(\lambda_2)})$	$\dots$	$(v_{r,1}, \dots, v_{r,m(\lambda_r)})$
<u>Eigenvectors</u> with e-value λ_1	<u>all eigenvectors</u>		

Since $\sum_{i=1}^r m(\lambda_i) = n = \deg \chi_A(x)$

these combine to give n vectors in $\mathbb{C}^n$

These give an eigenbasis

